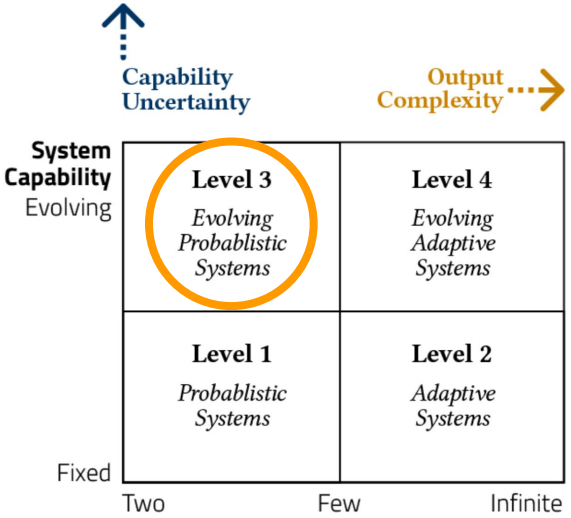
Confidence-Aware Demand and Inventory Planning Tool Interaction Walkthrough

Team No : 10

Jeena Shrestha, Vishnu Purohitham, Tarakaram Yadalla, Gil Benezer

Demand Modeling For Stockers and Planners

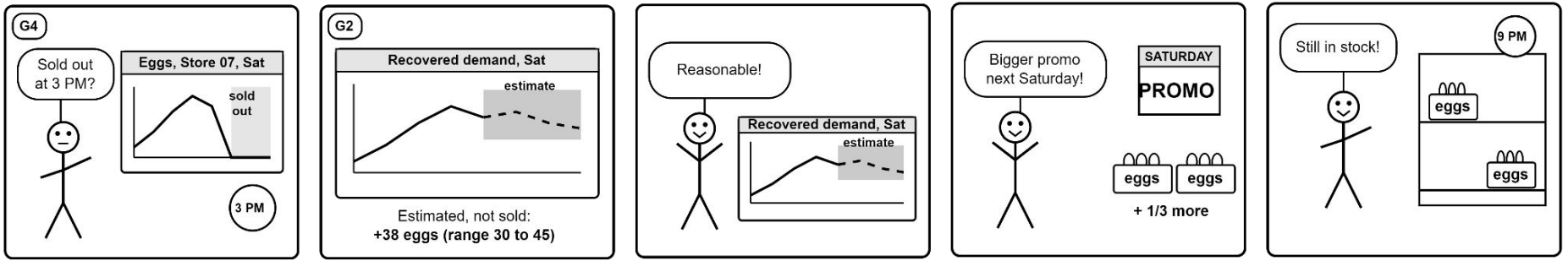
- Two Main AI Features
  - Model that imputes missing historical demand
  - Model that uses imputed and actual demand data to forecast
- Both Yang Level 3
- Two stakeholders
  - Imputation model: Demand Planner
  - Forecast model: Inventory Stocker





Demand Recovery Storyboard A

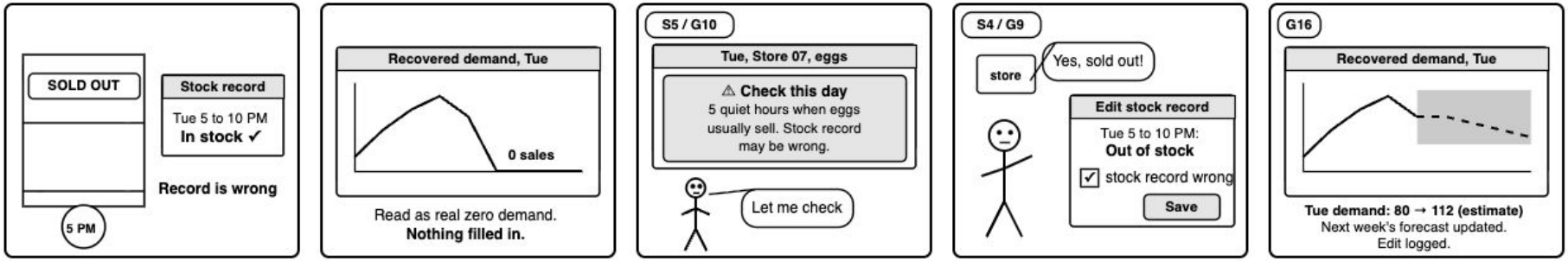
(Reader: Demand planner /Inventory Stocker)



Person acts on the output in panel 3. Panels 1 and 2 show what the system can do and how well.
Nothing goes wrong here, so there is no recovery panel.

Demand Recovery Storyboard B

(Reader: Demand planner /Inventory Stocker)



Shelf empty, record says in stock | **Model reads it as zero demand** | **System doubts the day, asks Maya to check** | **Maya corrects the record** | **Recovery reruns, edit is logged**

Datasheet entry behind this error (Errors, noise, or redundancies): **"Stockout labels are described as verified, but the verification method is unknown."**

System shows doubt in panel 3 (S5 / G10); Maya recovers and acts on the output in panel 4 (S4 / G9); panel 5 shows the consequence (G16).

Forecasting Model Storyboard A

Reader: Inventory Stocker

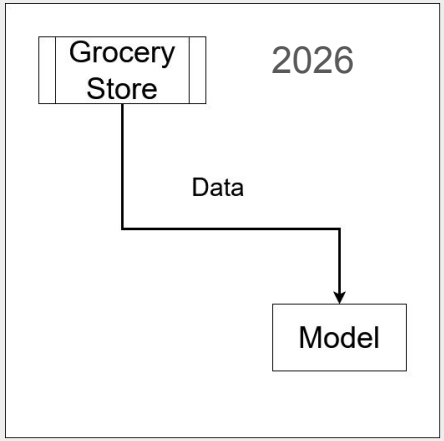
An automatic data collection pipeline feeds grocery store data into the forecasting model.

The model then outputs estimated egg demand over the next 7 days with 95% confidence intervals.

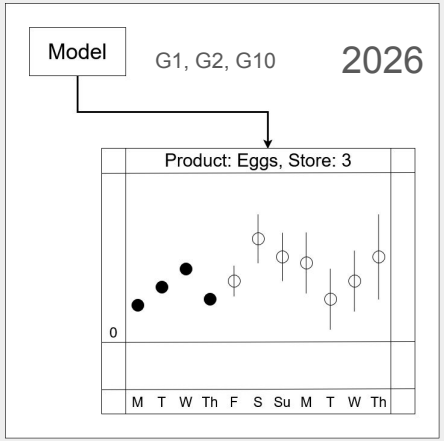
Independent covariate data is shown to explain estimates.

An inventory stocker trusts in the forecasts and buys appropriate amount of eggs.

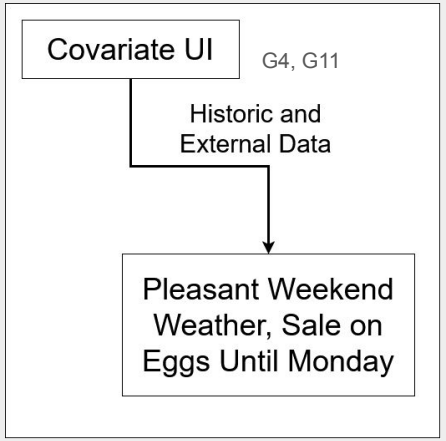
The business receives a profit.



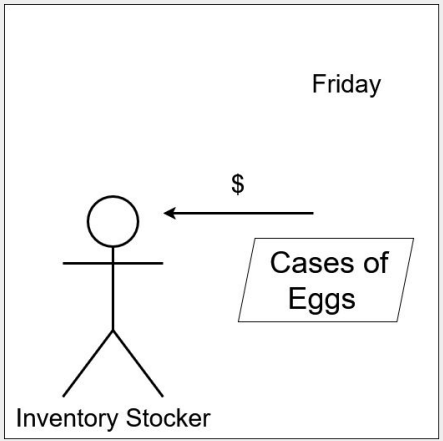
Data Automatically Collected and Fed to Model



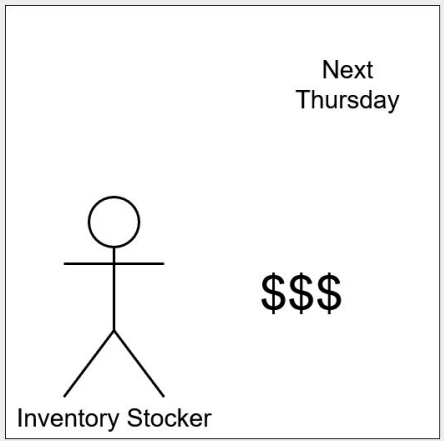
Model Outputs Needed Product and 95% Confidence Intervals



UI Has Graphical Output for Covariates



Stocker Buys Appropriate Amount of Eggs



Profit

Revisions

- First panel shows data feeding into model instead of disconnected model output
- Actors are only shown in panels where actions are taken
- Graphical interface shown instead of text (client input)
- Explicit interpretability metrics (SHAP, PI coverage percent, historical instance count) were removed from panel 3 (client input)

Forecasting Model Storyboard B1

Reader: Inventory Stocker

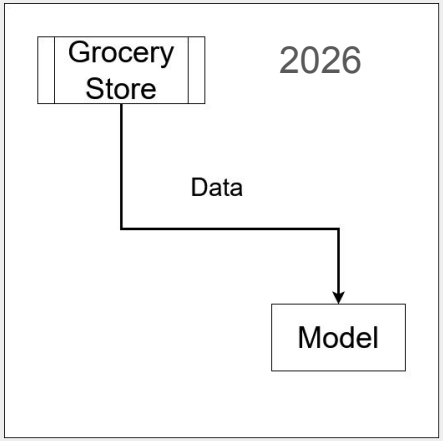
The automatic data collection pipeline from before feeds sparse data for quail eggs into the model.

The demand forecasts are untrustworthy because they are near or below zero.

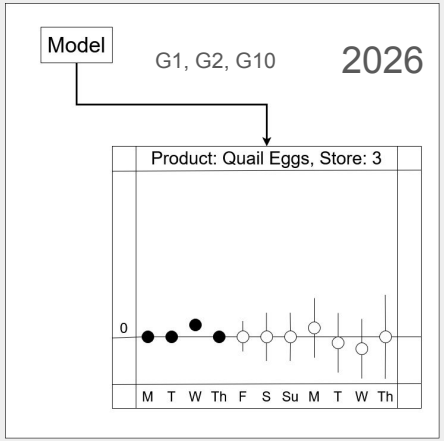
Covariates also flag and explain that the predictions are erroneous.

A stocker goes to the manual override interface and changes the underlying data.

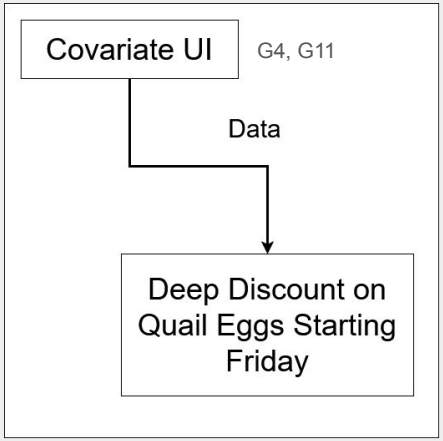
The stocker then orders based on the values entered in the manual override.



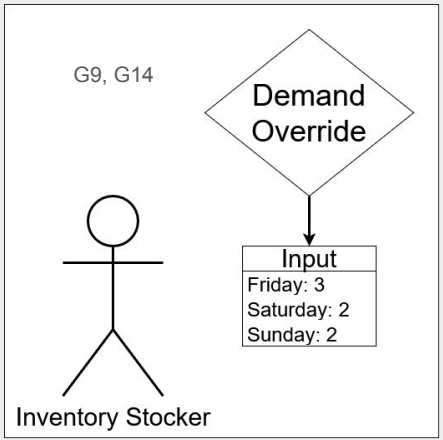
Data Fed to Model, But Data for Rare Product is Sparse



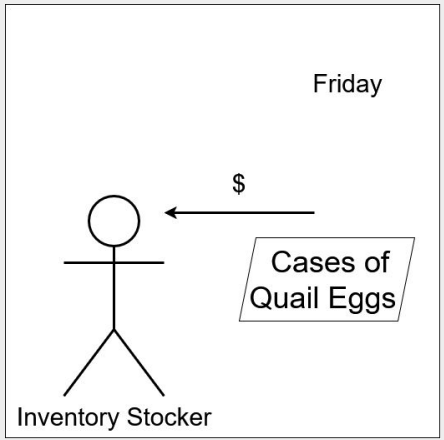
Model Outputs Nonsensical Demand Given Extrapolation



Covariates and Demand Estimates Conflict



Stocker Estimates Demand with Domain Knowledge



Stocker Orders Amount They Estimated

Revisions

- First panel shows data feeding into model instead of disconnected model output
- Actors are only shown in panels where actions are taken
- Graphical interface shown instead of text (client input)
- Explicit interpretability metrics (SHAP, PI coverage percent, historical instance count) were removed from panel 3 (client input)
- Recovery explicitly shown through human relying on domain knowledge

Forecasting Model Storyboard B2

Reader: Inventory Stocker

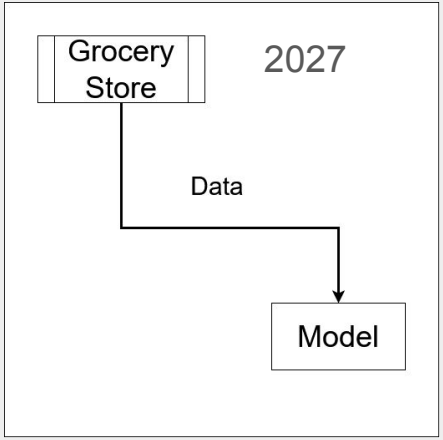
The automatic data collection pipeline from before feeds old data for eggs into the model.

The demand forecasts are not useful because the model has not been re-trained for a year and the distribution has shifted.

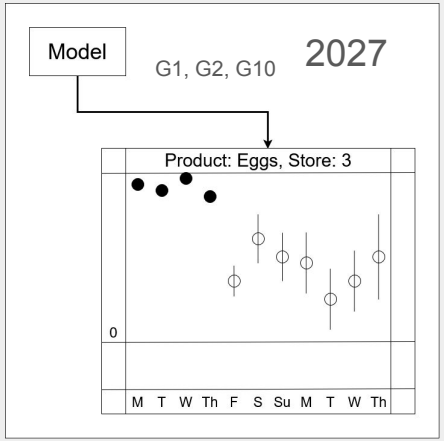
Covariates do not flag issues, but an experienced stocker notices shift.

A stocker edits the underlying forecast data to adjust the output.

The stocker then orders based on their domain expertise.



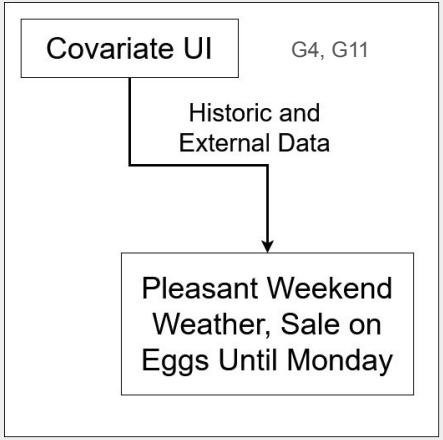
Data is Fed to the Model, but Distribution Has Shifted



● actual sales

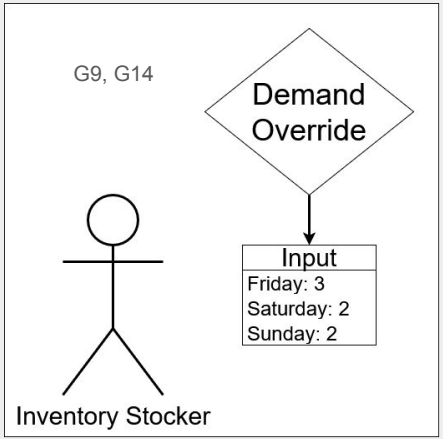
○ forecast with range.

Pre-Trained Model Not Adapted to New Distribution

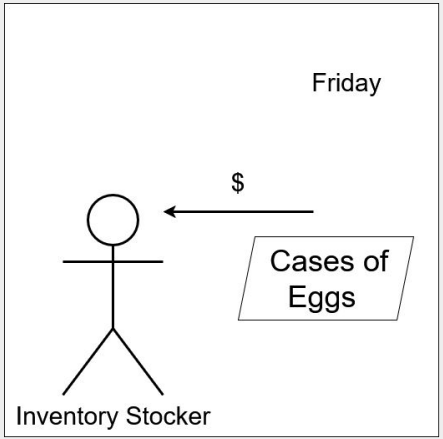


Similar Covariates, Different Temporal Context

Datasheet entry behind the error: the training data covers only the till the year 2026, so later shifts are never seen.



Stocker Edits Underlying Data Based on Domain Expertise



Stocker Orders based on Domain Expertise

Backup Slides

Forecasting Model Storyboard A

Reader: Inventory Manager

1. On Monday, Sam opens next week's egg forecast for Store 07, which notes that it now includes Maya's Tuesday correction.

2. The forecast shows a narrow daily range with high confidence, and last month's ranges held on 27 of 30 days.

3. Sam sets the egg order and safety stock from the top of the range.

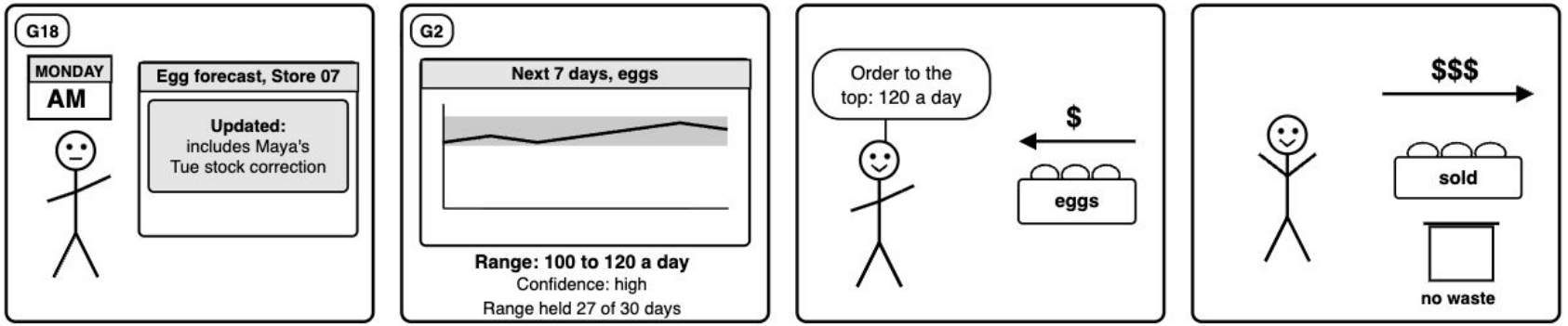
4. The eggs sell steadily through the week, with almost nothing thrown away.

1. On Monday, Sam opens next week's egg forecast for Store 07, which notes that it now includes Maya's Tuesday correction.

2. The forecast shows a narrow daily range with high confidence, and last month's ranges held on 27 of 30 days.

3. Sam sets the egg order and safety stock from the top of the range.

4. The eggs sell steadily through the week, with almost nothing thrown away.



Forecast updated with a correction | **Narrow range, high confidence** | **Sam orders from the range** | **Sold through, little waste**

Person acts on the output in panel 3. Panel 1 shows the change from Storyboard 1B (G18); panel 2 shows how well the system does (G2). Nothing goes wrong here, so there is no recovery panel; Storyboard 2B is its pair.

Forecasting Model Storyboard B

Reader: Inventory Manager

1. On Monday, Sam opens next week's egg forecast for Store 1. The forecast for National Day week shows low egg demand with a narrow range and high confidence, but the March to June training data never contained that holiday.

2. Sam trusts the forecast and orders fewer eggs than usual.

3. On the holiday weekend, the shelf is empty by noon and customers leave without eggs.

4. After demand lands above the range two days in a row, the system lowers the label to "check" and explains that it has never seen this holiday.

5. Sam adds the holiday effect to the data, raises the rest of the week's order, and flags the model for retraining.

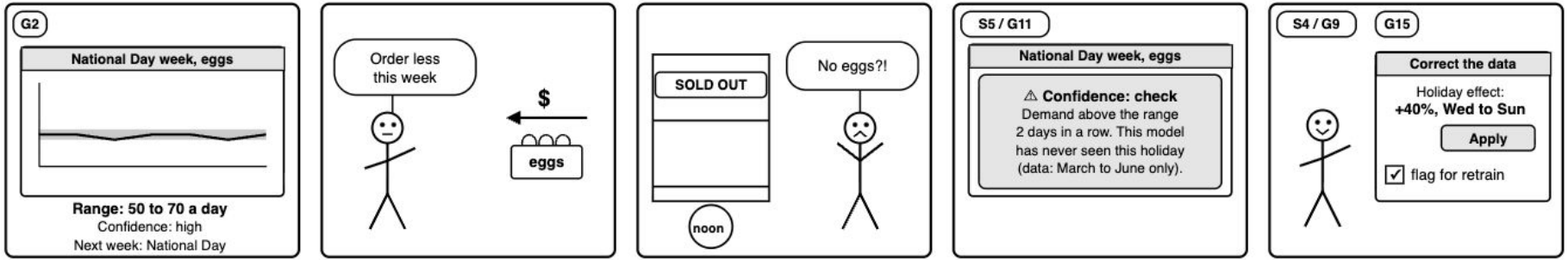
1. The forecast for National Day week shows low egg demand with a narrow range and high confidence, but the March to June training data never contained that holiday.

2. Sam trusts the forecast and orders fewer eggs than usual.

3. On the holiday weekend, the shelf is empty by noon and customers leave without eggs.

4. After demand lands above the range two days in a row, the system lowers the label to "check" and explains that it has never seen this holiday.

5. Sam adds the holiday effect to the data, raises the rest of the week's order, and flags the model for retraining.



Datasheet entry behind this error (Timeframe): "A three-month observation period, March to June 2024." Holidays outside that window were never in training.

Sam acts on the output in panel 2; the system shows doubt and explains itself in panel 4 (S5 / G11); Sam recovers in panel 5 (S4 / G9).
Change over time (Level 3): the retrain flag (G15) feeds the next training run, so later forecasts learn this holiday.

1 Search product

Product ID

SKU-104

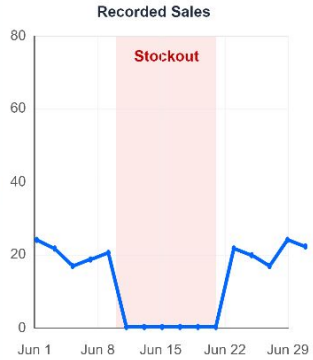
Store

Store 12



Planner opens a store–product record.

2 Find stockout



Sales are zero during stockout, so true demand is missing

3 Run imputation

Imputation model

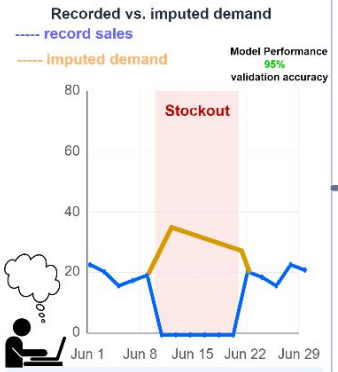
Inputs

- ✓ Historical sales
- ✓ Stockout dates
- ✓ Store/product context

Run model

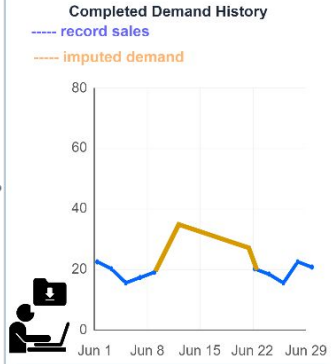
System estimates missing demand for the stockout period

4 Review Estimate



Planner compares estimated demand with recorded sales.

5 Save Corrected History



Planner accepts the estimate and saves the completed demand history.